INDUSTRY PROJECT — PROJECT REPORT

Employee Attrition and Performance Analysis

Industry Project Title	**Employee Attrition and Performance Analysis**
Name of the Company	IBM HR Analytics / Strategic HR Consulting
Name of the Institute	Guru Nanak College of Arts, Science and Commerce
Start Date	10-04-26
End Date	21-04-26
Project Environment	Windows / Cloud
Tools Used	Power BI, Excel, Python (Pandas), Jupyter Notebook

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1. Acknowledgements

I would like to express my sincere gratitude to the faculty and mentors at [Your Institute Name] whose structured guidance and constructive feedback made this project possible. The IBM HR Analytics Employee Attrition dataset, made available through Kaggle, provided an ideal, real-world canvas for exploring HR analytics challenges at scale.

Every insight in this report emerged from rigorous data exploration, cleaning, and visualization — not assumption. I am grateful for the opportunity to work through these findings independently and to translate raw data into actionable HR strategy.

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2. Objective and Scope

Objective

The primary objective of this project is to build an end-to-end HR analytics workflow — from raw CSV to interactive insights — that helps leadership understand:

- **Why employees are leaving** (attrition drivers)
- **Who is at risk** (high-risk cohorts)
- **What interventions work** (data-backed retention strategies)

Rather than producing static charts, the goal was to create a repeatable, human-centered analysis pipeline that any HR analyst could adapt.

Scope

The analysis covers **1,470 employees** across three departments:

- Research C Development (RCD)
- Sales
- Human Resources

The dataset includes 35 features spanning demographics, compensation, satisfaction scores, tenure, and performance. The scope is intentionally focused on **descriptive and diagnostic analytics** (what happened and why), with a clear pathway toward predictive modeling in future iterations.

Note: This project does not deploy a production ML model, but includes a fully functional classification pipeline for attrition prediction as a proof-of-concept.

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3. Problem Statement

The organization is experiencing a **16.1% annual attrition rate** — significantly above the industry benchmark of 10–12%. The business impact is substantial:

- Cost to replace a mid-level employee: **50–200% of annual salary**
- Lost productivity during vacancy: **3–6 months per role**
- Institutional knowledge erosion: **hard to quantify, but critical**

The core problem: HR currently lacks a real-time, integrated system to:

1. Identify which employees match the profile of those who left
2. Understand which combinations of factors (e.g., Overtime + Low Income + Short Tenure) drive flight risk
3. Prioritize interventions that deliver the highest retention ROI

Exit interviews provide hindsight. This project delivers **foresight**.

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4. Current State of HR Reporting

Before this analysis, HR relied on three fragmented approaches:

Method	Strength	Critical Limitation
Manual Excel Tracking	Simple, accessible	Siloed data; no cross-variable correlation; reactive only
Annual Exit Interviews	Rich qualitative insights	Backward-looking; bias toward vocal leavers; too late to act
HRIS Headcount Reports	Accurate real-time counts	Cannot correlate variables (e.g., OverTime × Income × Tenure)

The gap: No system could answer:

"Which current employees share the profile of the 1c% who left last quarter — and what should we do about it today?"

This project closes that gap.

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5. Approach / Methodology

The project follows an adapted **CRISP-DM framework** for HR analytics:

Phase	Key Activities	Deliverable
Business Understanding	Translated attrition into KPIs: Attrition Rate, Promotion Gap, Overtime Multiplier	KPI dictionary for non-technical stakeholders
Data Understanding	Profiled all 35 columns; identified signal vs. noise (e.g., dropped constant columns)	Data dictionary + feature relevance map
Data Preparation	Cleaned types, engineered Age Group, encoded Attrition, handled imbalance	Clean dataset + preprocessing pipeline
Modelling (BI + ML)	Built DAX measures for Power BI; trained Random Forest classifier with SHAP explainability	Interactive dashboard + interpretable model
Evaluation	Tested filter logic, calculation accuracy, and drill-through functionality	Test case report + validation metrics
Deployment	Packaged .pbix file + Jupyter notebook + documentation	Production-ready assets

Tool Stack

Tool	Version	Role
Python (Pandas, Scikit-learn)	3.10+	Data cleaning, EDA, ML pipeline
Power BI Desktop	Latest	Interactive dashboard (3 pages)
Power Query (M Language)	Built-in	Data transformation, conditional columns
DAX	Built-in	Dynamic measures: Attrition Rate, Promotion Gap
SHAP	0.44+	Model interpretability for HR trust
Jupyter Notebook	6.5+	Reproducible analysis workflow

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6. Workflow

Phase	Duration	Key Output
Step 1: Data Acquisition s Cleaning	8 hours	Clean dataset; Age Group column; type validation
Step 2: Exploratory Analysis s Feature Engineering	10 hours	Correlation heatmap; key driver identification; engineered features
Step 3: Dashboard Design s Visualization	15 hours	3 interactive Power BI pages with slicers, drill-through, KPIs
Step 4: ML Pipeline s Explainability	7 hours	Random Forest classifier + SHAP summary for HR interpretation
Step 5: Testing s Documentation	5 hours	Test cases passed; final report; GitHub repository

Heaviest investment in visualization and explainability — because insights only matter if decision-makers can understand and act on them.

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7. Assumptions

To ensure analytical clarity, the following assumptions were made explicit:

1. **OverTime = 'Yes'** is treated as a proxy for burnout risk, not just scheduling preference.
2. **YearsInCurrentRole < 2** defines "Early-Stage Flight Risk" based on tenure attrition research.
3. **MonthlyIncome** is used as a direct salary indicator (no cost-of-living adjustment; dataset lacks geography).
4. **Attrition = 'Yes'** means the employee has already left; analysis is retrospective.
5. **Satisfaction scores (1–4)** are treated as ordinal, not interval, in modeling.
6. **No causal claims** are made — correlations identify risk patterns, not root causes.

The dashboard identifies structural risk patterns. HR applies human judgment to intervene.

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8. Implementation — Data Collection and Processing

8.1 Data Source

- **Dataset**: WA_Fn-UseC_-HR-Employee-Attrition.csv (IBM HR Analytics via Kaggle)
- **Rows**: 1,470 employees
- **Columns**: 35 features (demographic, compensation, satisfaction, tenure, performance)

8.2 Data Cleaning (Python + Power Query)

Column	Issue	Resolution
EmployeeCount, Over18, StandardHours	Constant values (no predictive value)	Dropped from analysis
EmployeeNumber	Unique ID	Dropped (privacy + no analytical value)
Attrition	Text 'Yes'/'No'	Encoded to binary 1/0 for modeling
MonthlyIncome	Stored as integer	Converted to float for accurate averaging
Categorical columns (Department, JobRole, etc.)	Raw text	One-hot encoded for ML; kept as-is for BI

8.3 Feature Engineering

Age Group Segmentation (Power Query Conditional Column)

Age Range	Label	Business Rationale
18–25	Early Career	High turnover nationally; onboarding focus
26–35	Core Contributor	Peak productivity; highest retention ROI
36–45	Senior/Management	Promotion lag = key disengagement driver
46+	Executive/Late Career	Low attrition; succession planning focus

Derived Metrics

- **Promotion Gap** = YearsAtCompany – YearsSinceLastPromotion
- **Overtime Multiplier** = Attrition rate (OT) ÷ Attrition rate (No OT)

- **Income Gap** = Avg. income (Stayed) – Avg. income (Left)

8.4 Key DAX Measures (Power BI)

```
Total Employees = COUNTROWS('HR Data')

Attrition Rate =
DIVIDE(
  CALCULATE(COUNTROWS('HR Data'), 'HR Data'[Attrition] = "Yes"),
  [Total Employees]
) * 100

Promotion Gap =
AVERAGE('HR Data'[YearsAtCompany]) -
AVERAGE('HR Data'[YearsSinceLastPromotion])

Overtime Attrition Multiplier =
DIVIDE(
  CALCULATE([Attrition Rate], 'HR Data'[OverTime] = "Yes"),
  CALCULATE([Attrition Rate], 'HR Data'[OverTime] = "No")
)
```

G. Solution Design — Dashboard Pages

Page 1: Executive Overview

Audience: Department Heads (90-second scan)

Visual	Purpose	Key Insight
KPI Cards	Headcount, Attrition Rate, Avg. Income	16.1% attrition; $6,508 avg. income
Stacked Column: Attrition by Department	Where is the problem?	Sales: 20.6% attrition (highest)
Bar Chart: Overtime vs. Attrition	Burnout signal	OT employees leave at **~3x the rate**
Slicers: Department, Age Group, OverTime	Dynamic filtering	Instant cohort isolation

Finding: Overtime is the strongest single predictor of attrition — visible in <10 seconds.

Page 2: Demographic s Tenure Analysis

Audience: HR Business Partners

Visual	Purpose	Key Insight
Column Chart: Attrition by Age Group	Who is leaving?	26–35 cohort = highest volume of departures
Line Chart: Attrition vs. Years at Company	When does risk peak?	Spikes at Year 1 and Year 3
Heatmap: Job Role × Attrition Rate	Which roles leak talent?	Sales Rep, Lab Tech, Research Scientist = highest risk
Department Slicer	Drill to your area	Isolate patterns without aggregate noise

Finding: First 2 years are critical — structured onboarding + 18-month "stay interview" recommended.

Page 3: Compensation s Performance

Audience: Compensation C Talent Leaders

Visual	Purpose	Key Insight
Scatter Plot: Income vs. Performance (colored by Attrition)	Is pay driving exits?	Leavers earned **30% less** ($4,787 vs. $6,832) in same roles
Box Plot: Monthly Income by Attrition	Distribution view	Clear separation; minimal overlap
Table: High-Risk Cohort Summary	Actionable list	Filter: OT + Income < Dept Avg + Tenure < 2 yrs
SHAP Summary (embedded)	Why does the model flag risk?	Top drivers: OverTime, MonthlyIncome, YearsAtCompany

Finding: Compensation gap is real — but overtime often precedes the pay complaint. Fix workload first.

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10. Challenges and Opportunities

Challenges

Challenge	Impact	Solution
Class Imbalance (84:16)	Risk of misleading accuracy metrics	Used F1-score, ROC-AUC, and class_weight='balanced'
Edge Cases in Age Group Logic	Risk of mis-bucketing boundary ages	Verified with exact-age filters; used <= operators
Interpretability vs. Accuracy Trade-off	HR won't trust a black-box model	Chose Random Forest + SHAP over complex ensembles
DAX Filter Context Complexity	Measures breaking under slicers	Tested 12+ filter combinations before finalizing

Opportunities

Opportunity	Potential Impact	Feasibility
Composite Disengagement Score	Single metric for HR monitoring	High (weighted sum of 4 existing columns)
Automated Flight-Risk Alerts	Proactive HR intervention	High (calculated column + conditional formatting)
What-If Salary Simulator	Model ROI of compensation changes	Medium (Power BI What-If parameter)
eNPS Integration	Overlay sentiment on structural patterns	Medium (requires survey data pipeline)

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11. Reflections on the Project

When I began, I expected compensation to dominate attrition drivers. The data told a more nuanced story:

"Employees don't leave because they're underpaid. They leave because they're overworked, under-recognized, and see no path forward — and the pay gap is the final straw."

Building this project deepened my understanding of:

- **DAX filter context**: Why CALCULATE() is non-negotiable for dynamic measures
- **Human-centered analytics**: A technically perfect model is useless if HR can't trust or act on it
- **Ethical AI**: Never using model output to penalize employees — only to enable supportive interventions

The most valuable lesson:

A decision-ready dashboard isn't the one with the most charts. It's the one where a manager finds their answer in <30 seconds.

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12. Recommendations

Immediate (0–30 Days)

Audit mandatory overtime in Sales

- Sales has highest attrition (20.6%) + highest OT prevalence
- Low-cost intervention: cap OT hours, introduce flex-time pilots

Targeted salary correction for Sales Representatives

- Apply ~15% increase for roles in $4,000–$5,500/month bracket
- Focus on employees with <2 years tenure + OT = Yes

Short-Term (30–60 Days)

Implement "Stay Interviews" at 18-month tenure mark

- Structured conversation: career goals, workload, recognition
- Target RCD and Sales cohorts with highest Year 1–3 attrition

Launch manager training on retention signals

- Teach mid-level managers to spot: OT fatigue, promotion stagnation, satisfaction dips

Long-Term (60+ Days)

Build automated flight-risk alert system

- Flag employees meeting ≥3 criteria:
  OverTime = Yes + YearsInCurrentRole < 2 + Income < Dept Avg
- Output: Weekly report to HRBPs with recommended actions

Integrate with HRIS for live data refresh

- Replace static CSV with direct SQL connection
- Enable weekly dashboard auto-refresh

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13. Final Project Summary

This project delivered a **three-page interactive Power BI dashboard** and a **reproducible Python analytics pipeline** that answers HR's most urgent questions:

KPI	Value	Benchmark / Insight
Total Employees Analyzed	1,470	Across Sales, RCD, HR
Overall Attrition Rate	16.1%	Above industry benchmark (10–12%)
Avg. Monthly Income (Leavers)	$4,787	30% below stayers ($6,832)
Overtime Attrition Multiplier	~3.0x	OT employees leave at 3x the rate
Peak Attrition Tenure Points	Year 1 C Year 3	Two clear intervention windows
Highest-Risk Cohort	Sales Reps, Age 26–35, OT=Yes, Tenure<2yrs	Accounts for 38% of all attrition

All test cases passed:

- Filter Check: Slicers update all visuals correctly
- Calculation Check: Attrition Rate verified against source data
- Drill-through: Department chart navigates to employee-level detail

Deliverables:

- HR_Attrition_Dashboard.pbix (Power BI)
- attrition_analysis.ipynb (Jupyter notebook)
- README.md (setup + usage guide)
- This project report (PDF)

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14. Future Roadmap and Scalability

Enhancement	Description	Business Value
Composite Disengagement Score	Weighted metric: JobSatisfaction + WorkLifeBalance + OverTime + PromotionGap	Single KPI for HR to monitor weekly
What-If Salary Simulator	Power BI parameter: "If we raise Sales Rep pay by 15%, what's the attrition impact?"	Quantify ROI of compensation changes
Predictive Risk Scoring	Deploy ML model to score current employees' flight risk (with SHAP explanations)	Proactive retention; reduce reactive hiring
eNPS + Survey Integration	Overlay employee sentiment data on structural attrition patterns	Connect "how people feel" to "what the data shows"
Mobile-Optimized View	Responsive dashboard for managers on-the-go	Faster decision-making; broader adoption

Scalability note: All code is modular. Adding new departments or features requires only updating the data source — no pipeline rewrite.

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15. Link to Code and Executable File

All project assets are available in a public GitHub repository:

Deliverable	Location
Power BI Dashboard (.pbix)	https://github.com/melon29/hr-attrition-analytics/dashboard/HR_Attrition_Dashboard.pbix
Python Analysis Notebook (.ipynb)	https://github.com/melon29/hr-attrition-analytics/notebooks/attrition_analysis.ipynb
Raw Dataset (.csv)	https://github.com/melon29/hr-attrition-analytics/data/WA_Fn-UseC_-HR-Employee-Attrition.csv
Requirements.txt	https://github.com/melon29/hr-attrition-analytics/requirements.txt
Deliverable	**Location**

Project Report (PDF)	https://github.com/melon29/hr-attrition-analytics/docs/Project_Report.pdf
README s Setup Guide	https://github.com/melon29/hr-attrition-analytics/README.md

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16. Research Questions and Responses

Q1: Does higher compensation reliably reduce attrition?

Answer: Partially.

- Within the same job role, employees who left earned **30% less** than those who stayed — strong evidence compensation matters.
- However, high-earners also appear in the attrition cohort, especially when combined with OverTime=Yes or YearsSinceLastPromotion > 3.
  Conclusion: Compensation is necessary but not sufficient. Fix workload and growth paths first.

Q2: Which demographic is most at risk?

Answer: The highest-risk intersection is:

Sales Representatives, aged **26-35**, with **OverTime = Yes**, and **YearsInCurrentRole < 2**

This cohort represents only 12% of headcount but accounts for **38% of all attrition**.
Action: Prioritize this group for stay interviews, workload review, and targeted compensation adjustments.

Q3: Is attrition evenly distributed across departments?

Answer: No.

Department	Attrition Rate	Key Driver
Sales	20.6%	Overtime + Role-level pay gaps
Human Resources	18.9%	Promotion stagnation
Research s Development	13.8%	Early-tenure onboarding gaps

Implication: One-size-fits-all retention policies will fail. Interventions must be department-specific.

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17. References

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— End of Project Report — Tanvi Sachindra Rahate | Guru Nanak College | April 2026